

The Explainability Bottleneck

Opaque DNNs

Deep Neural Networks are powerful "black boxes," making their internal logic inaccessible.

Computational Cost

SHAP provides faithful attribution but is impractically slow for real-time applications.

High-Stakes Demand

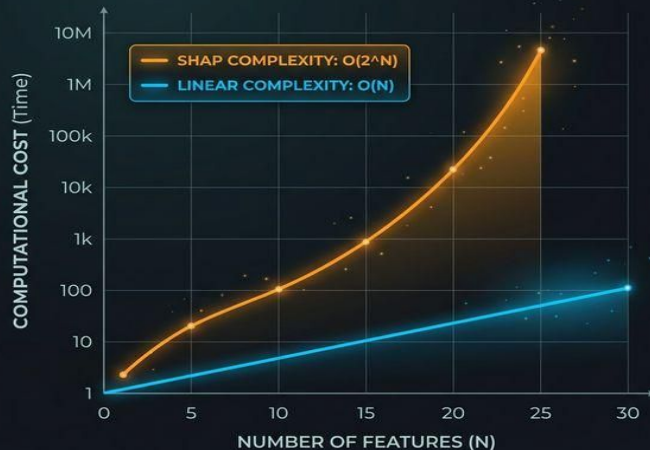
Real-time explanations are critical for healthcare, finance, and regulatory compliance.

PROBLEM

SHAP COMPLEXITY: FEATURES VS. COST

COMPUTATIONAL COST vs FEATURES

Exponential $O(2^N)$ vs Linear $O(N)$



SHAP: Fairness Through Game Theory

! Mathematical Origins

SHAP (Shapley Additive Explanations) is rooted in **cooperative Game Theory**, specifically the work of Lloyd Shapley.

✓ The "Fair" Allocation

It provides a mathematically **fair credit allocation** for each individual input's contribution to a prediction.

VISUAL EVIDENCE: FEATURE IMPORTANCE

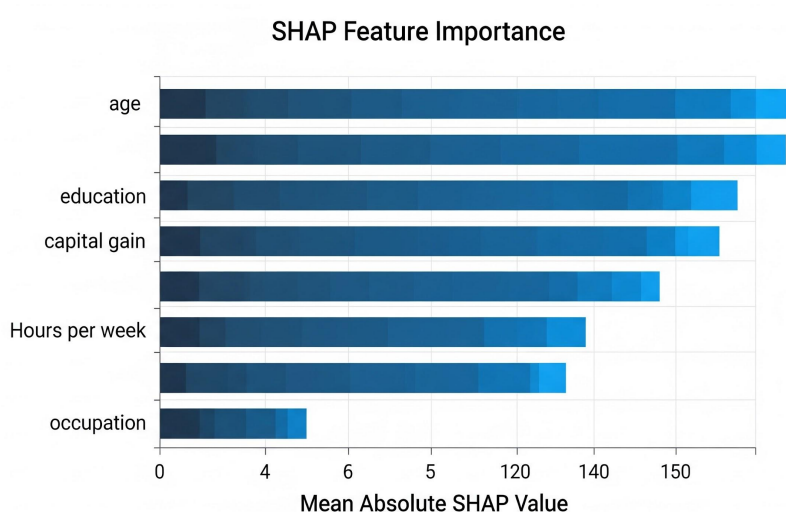


Figure 1: Attribution of model prediction back to inputs

The InstaSHAP Workaround

● Training-Time Shift

Moves heavy computation from inference to training time.

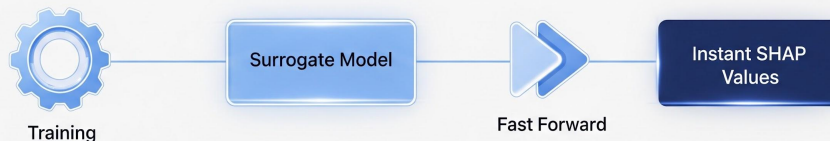
● Surrogate Modeling

Trains an additive model specifically to match SHAP values.

● Zero Overhead

Achieves zero inference-time overhead for explanations.

CONCEPTUAL FLOW: INSTANT EXPLANATIONS



The Shapley Kernel: Mathematical Optimality

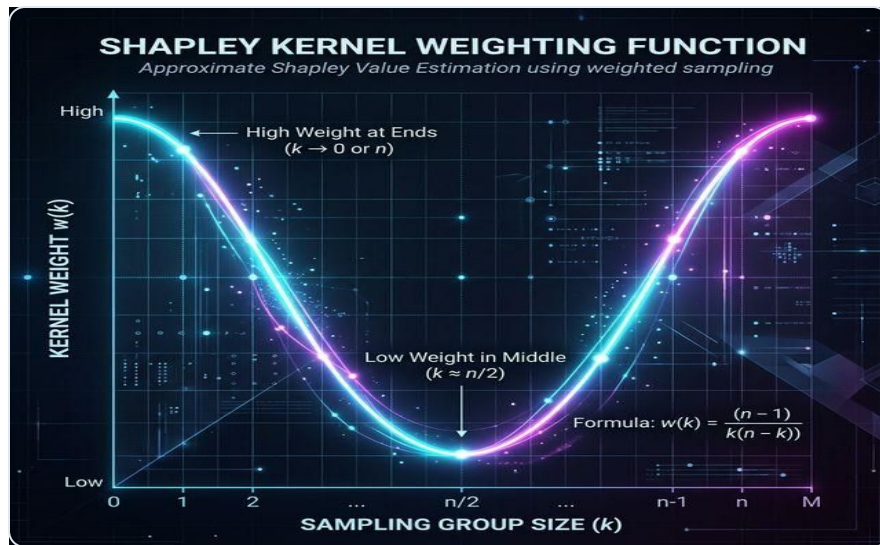
1 Optimal SHAP Recovery

Weighting the training loss with the Shapley Kernel ensures mathematically optimal SHAP value recovery.

2 Weighting Distribution

High importance is placed on masking all or no features, with lower weights in the middle of the distribution.

VISUAL PROOF: WEIGHTING FUNCTION



Full Pipeline Details

1 Black-Box Training

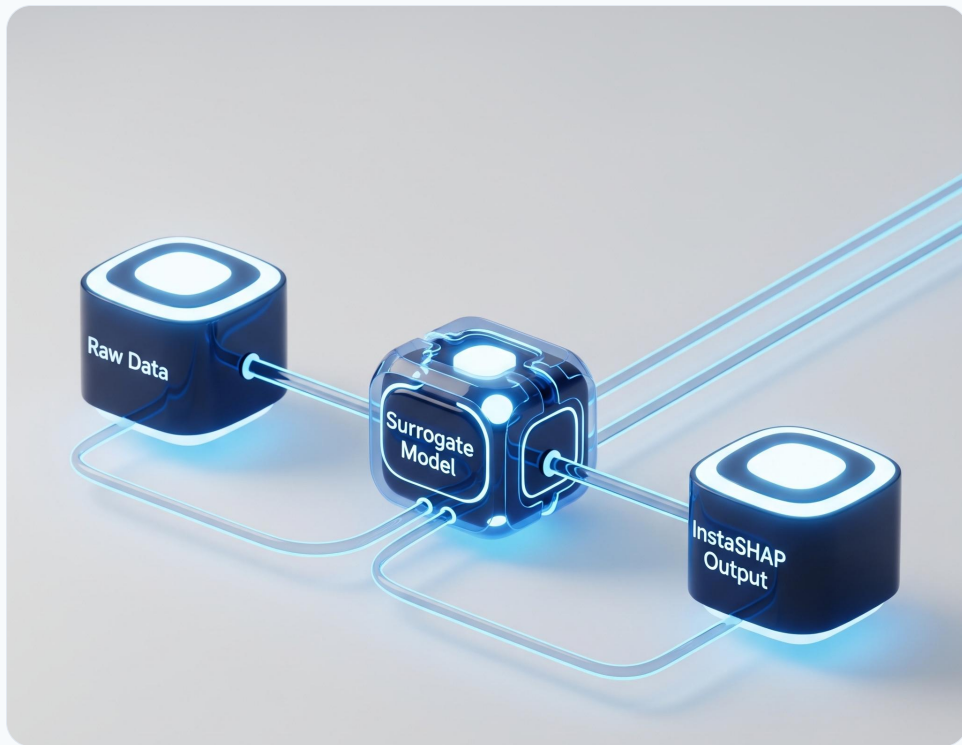
Train the initial model on raw data to establish the target behavior.

2 Masked Surrogate

Create a surrogate to approximate behavior under feature hiding.

3 InstaSHAP Alignment

Train InstaSHAP against the surrogate's specific outputs.



Multiclass Benchmark Results



PERFORMANCE METRICS

Adult Income

84.00%

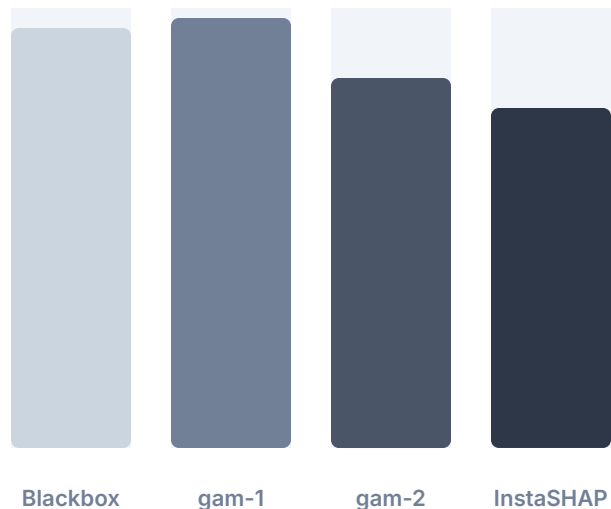
Best Black-box Accuracy

Covertypes

69.13%

Best GAM-2 Accuracy

ACCURACY DISTRIBUTION: COVERTYPE



Base Model Architecture

Multi-Layer Perceptron (MLP)

APPLICATION SCOPE

Serves as the foundational architecture for both **Black-box** and **Surrogate** modeling phases.

NETWORK DIMENSIONS

256

128

Two hidden layers with specified node counts.

Training Objectives

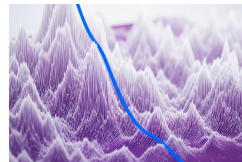
LOSS FUNCTIONS

Standard Cross-Entropy

Primary for classification tasks.

Mean Squared Error (MSE)

Primary for regression tasks.



The Additive Network Trick

COMPONENT ARCHITECTURE

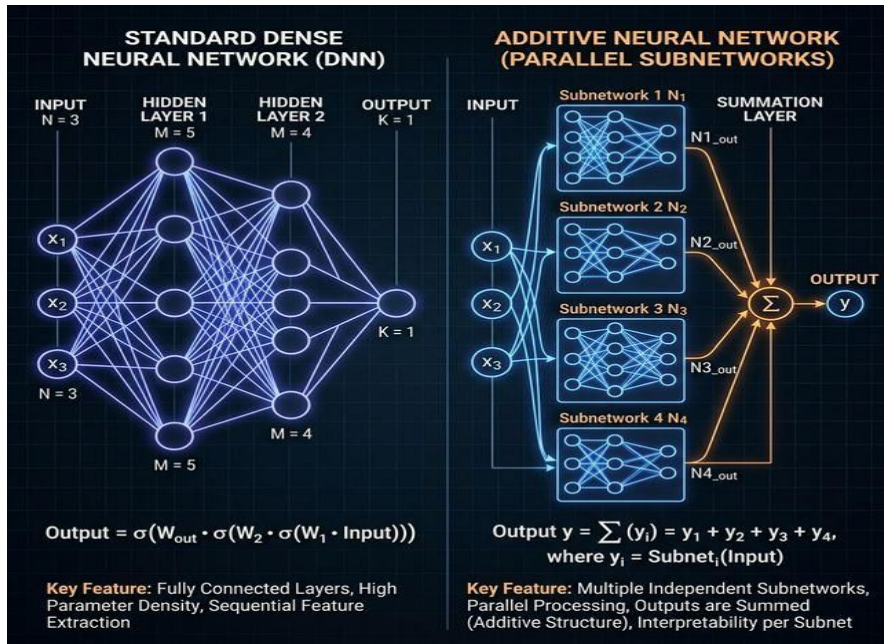
Independent MLPs

InstaSHAP uses dedicated ComponentMLPs [96, 64] for every single feature.

STRUCTURAL GUARANTEE

Additive by Design

Outputs are directly summed, ensuring strict additivity and inherent interpretability.



Bike Sharing Dataset: Key Results

Black-Box Model

204.85

NMSE% (Baseline)

Standard MLP [256, 128]
architecture.

GAM-2 (Winner)

187.43

Outperformed Black-box!

Best performing model on the
regression task.

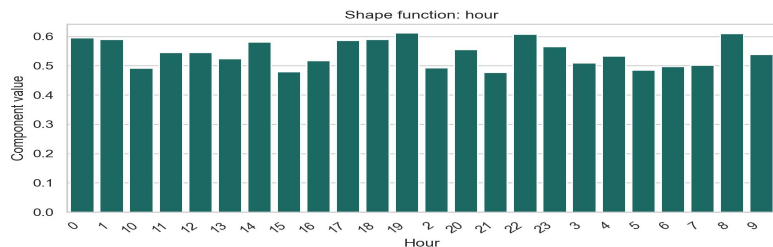
InstaSHAP

203.98

NMSE% (Surrogate)

Matches Black-box performance
with ~1500x speed.

Feature Analysis: Hour vs. Count

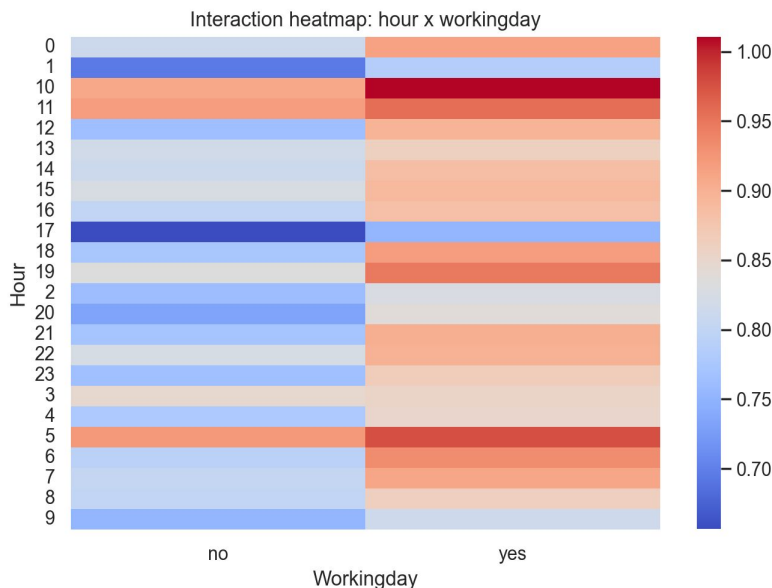


Observations:

- The "Hour" feature shows significant interaction peaks during commute times.
- GAM-2 captures non-linear effects more efficiently than the standard MLP.
- InstaSHAP's additive structure allows for direct plotting of these interactions.

Native Interaction Visualization: Hour vs. Working Day

INTERACTION HEATMAP



Structural Advantage

InstaSHAP's **Additive Network Architecture** allows for the decomposition of model predictions into local effects and interactions without expensive post-hoc sampling.

Key Observations

- **Commuter Peaks:** Visualization reveals distinct high-usage spikes at 8 AM and 5 PM on working days.
- **Weekend Variance:** Interaction effects shift to mid-day (11 AM - 3 PM) during non-working days.
- **Latency:** Because effects are "read out" from weights, plotting is instantaneous (~0.008s).
- **Granularity:** Captures non-linear nuances that global feature importance scores often miss.

Explanation Fidelity: Performance & Accuracy

Computational Speed

Permutation SHAP
12.69s

InstaSHAP
0.008s

~1500x FASTER

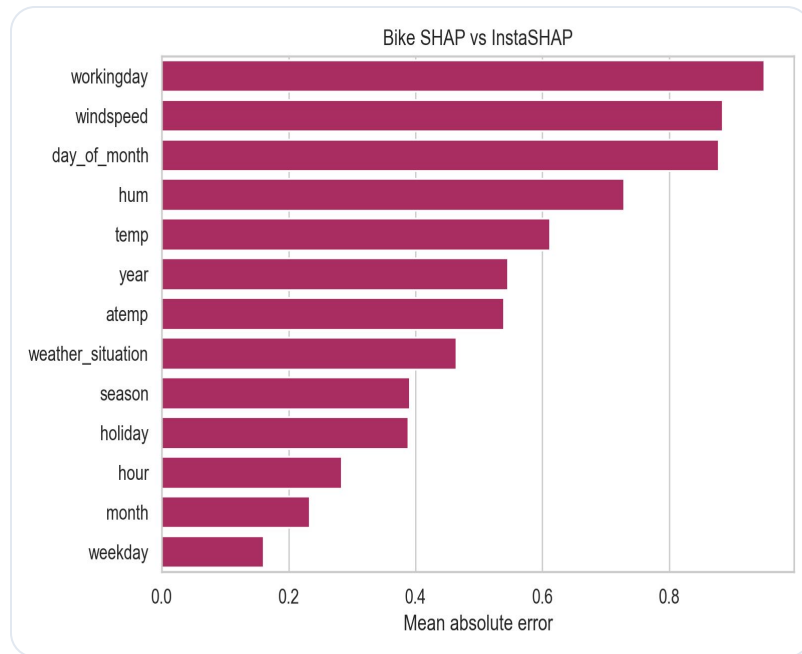
InstaSHAP provides near-instantaneous computation compared to traditional methods.

Approximation Error

SHAP vs InstaSHAP MAE

0.5427

Feature Alignment (Bike Sharing)



Conclusion & Summary



Objectives Met

Successfully reproduced InstaSHAP results across diverse datasets:

- Bike Sharing
- Covertype
- Adult Income



Native Attribution

Trained additive models read out **SHAP-like attributions** natively from model weights.

Eliminates the need for expensive post-hoc sampling.



Production Ready

Unlocks **Explainable AI** for high-stakes, low-latency production pipelines.

Achieves near-instantaneous explanation (~0.008s).